MIL-F-20412
4 December 1951

MILITARY SPECIFICATION

FUSE, SAFETY
(Bickford Type)

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments, and supersedes the following specification:

Army 50-69-1A
23 February 1934

This specification consists of this cover sheet and Specification 50-69-1A, dated 23 February 1934, attached hereto, without modification.

When a request for this specification is received by a supplying activity it will be necessary to attach this cover sheet to the pertinent specification before issue.

Copies of specifications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.

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No. 50-69-1A
February 23, 1934
Superseding
No. 50-69-1
June 16, 1930

U.S. ARMY SPECIFICATION

FUSE, SAFETY
(Bickford Type)

I. GENERAL SPECIFICATIONS.

1. The following United States Army Specifications and the applicable drawing, if any, of the issue in effect on the date of invitation for bids, shall form a part of this specification:

96-111-5, United States Army Specification for Matchhead.
100-2, United States Army Standard Specifications for Marking Shipments.

II TYPES.

1. This specification covers 3 types.

Type I. Burning time, 5.25 seconds per inch.
Type II. Burning time, 4.25 seconds per inch.
Type III. Burning time, 2.75 seconds per inch.

III. MATERIAL AND WORKMANSHIP.

1. The contractor shall furnish all materials required for the production, packing, marking, and shipping of the material contracted for, of the kind and type noted on the drawings and in detail specifications, except materials specifically excepted by the terms of the contract.

IV. GENERAL REQUIREMENTS.

1. Construction. -- The fuse shall consist of a thin train of powder tightly compressed in inner and outer wrappings of jute thread, cotton thread, or tape, with a suitable waterproofing material between them. The waterproofing material shall be of a grade that will be least affected by climatic conditions. The body of the fuse shall be uniform in cross-section and flexible throughout its entire length.

2. Diameter. -- The diameter of the fuse shall be as specified in the contract or as indicated on the applicable drawing.

3. Length. --The length of the fuse shall be as specified in the contract or as indicated on the applicable drawing, but in no case more than 50 lineal feet in one length.

4. Ends. -- The ends of each piece of fuse shall be cut perpendicular to the axis of the fuse, without fraying or other irregularities.

5. Starting paste. -- When so specified in the contract, one end of each piece shall be covered with matchhead, specification 96-111-5, or with the starting paste (priming cap) indicated on the applicable drawing. In case the contractor desires to use a matchhead, or starting paste, other than the one specified, he shall, prior to the use thereof, receive approval of the United States. The matchhead or starting mixture (priming cap) shall be firmly attached and shall be as indicated on the applicable drawing.

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6. Flame intensity. -- The flame intensity shall be such that the fuse will pass the test described in section VI, paragraph 4d.

V. DETAIL REQUIREMENTS.

1. Burning time.-- The burning time of the fuse shall be as shown in Table I:

TABLE I

Type	Time of burning, seconds per inch	
	Minimum	Maximum
I	5.0	5.5
II	4.0	4.5
III	2.5	3.0

TABLE II

Type	Color of center thread
I	Green thread visible in unprimed end.
II	Yellow thread visible in unprimed end.
III	Red thread visible in unprimed end.

VI METHODS OF INSPECTION, TESTS, ECT.

1. Inspection. -- Inspection shall be at the point of delivery unless otherwise specified in the contract.

2. Size of lots. -- a. When ordered in short pieces, 20,000 maximum.

b. When ordered in lineal feet, 6,000 maximum.

c. A smaller quantity of either pieces or lineal feet may constitute a lot at the discretion of the Government inspector.

3. Sampling. --a. When ordered in pieces, the Government inspector shall select 20 pieces of fuse from each lot for test.

b. When ordered in pieces, the contractor shall furnish from each lot of fuse for the waterproof test, 10 samples 12 inches long. These samples shall be primed on one end.

c. When ordered in lineal feet, the Government inspector shall select 10 reels or coils from each lot and cut three 12-inch pieces of fuse from each for test. When a lot contains less than 10 reels or coils, the Government inspector shall cut three 12-inch pieces of fuse from each for test. Dip one end of the sample fuse pieces in match-head paste, United States Army Specification 96-111-15 and allow to dry.

d. All specimens and samples shall be properly marked for identification.

e. The contractor shall always allow, without cost to the United States, sufficient additional material to furnish the test specimens or samples required, unless othe rwise provided in the contract. Any material required by the contractor for his own tests at the various stages of manufacture shall be in addition

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to that required by the United States for the test of acceptance, and in no case, unless by special authorization of the inspector, shall the contractor use for his own tests any of the material required by the United States.

f. Any additional specimens or samples required to establish the acceptability of the material represented, necessitated by the defects in the original specimens, or other like causes, shall be furnished at the expense of the contractor, and he shall bear all the expense of such additional tests.

4. Testing. -- a. Construction. -- The Government inspector shall make sufficient examination of the sample fuse pieces for defects in construction and to determine if the other requirements of the specification or applicable drawing are complied with. Reject the lot if defects in construction are found. Also reject the lot if any of the other requirements of the specification or applicable drawing are not met.

b. Waterproofness. -- Tie the two ends of each of 10 of the 12-inch sample fuse pieces so that each piece forms a loop. Then suspend each loop in a vessel of water to within 2 inches of the ends for 5 minutes. Remove from the water, unfasten ends, remove excess water, and ignite separately in a V-shaped trough by means of a red-hot iron or similar method applied to the primed ends of the fuze. Determine time of burning of each fuse piece by the use of a stop watch or equivalent method which is capable of recording correctly within one tenth of a second. Reject the lot, subject to retest, if any fuse piece fails to burn completely at the burning rate prescribed in Table I. Repeat the test once, using double the number of samples, in case a retest is requested by the contractor, if not more than two fuse pieces fail to meet the requirements of Table I provided no piece burns less than 0.5 second per inch below the computed burning time, using the minimum time of burning for the type being tested. Reject the lot if, in the retest, any fuse piece fails to burn completely at the burning rate prescribed in Table I.

c. Burning time. -- Ignite 10 of the sample fuse pieces separately in a V-shaped trough by means of a red-hot iron, or similar method, applied to the primed end of the fuse. Determine time of burning of each fuse piece by the use of a stop watch or equivalent method which is capable of recording correctly within one tenth of a second. Reject the lot, subject to retest, if the rate of burning of any piece fails to come within the limits specified in Table I. Repeat the test once, using double the number of samples, in case a retest is requested by the contractor, if not more than two sample fuse pieces fail to meet the requirements of Table I, provided no piece burns less than 0.5 second per inch below the computed burning time, using the minimum time of burning for the type being tested. Reject the lot if, in the retest, the rate of burning of any fuse piece fails to come within the limits specified in Table I.

d. Flame intensity. -- Form a train with 10 of the sample fuse pieces. Place them, with primed ends pointing the same direction, 1/2 inch apart, in a V-shaped trough. Ignite the primed end of the first piece. Reject the lot of fuse, subject to retest, if the flame from any fuse piece fails to ignite the adjacent fuse piece, or if an instantaneous or near-instantaneous action be obtained from any single fuse. Repeat the test once, using double the number of samples, in

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case a retest is requested by the contractor. Reject the lot if, in the retest, the flame from any fuse piece fails to ignite the adjacent fuse piece or if an instantaneous or near-instantaneous action be obtained from any single fuse.

Note. -- A misdirection of the flame from one fuse piece to the primed end of the next fuse will not be cause for rejection. The fuse pieces already burned through to the first fuse piece remaining unignited, may be replaced by fuses selected from the same lot and the test carried out as originally intended.

VII. PACKING AND MARKING.

1. Packaging. -- a. Pieces. -- Pack 1,000 pieces of fuse in a pasteboard box. The pasteboard box shall be lined with a waterproofed paper of suitable weight and thickness. The pasteboard box shall be stamped with the grade of fuse packed therein, and shall be wrapped in waterproofed paper.

b. Reels or coils. -- When ordered in lineal feet, the fuse shall be wound on reels or in coils of 50 feet each.

2. Packing. -- a. Pieces. -- Pack 20 pasteboard boxes in suitable metal liner. Hermetically seal the liner and provide with the usual tear strip and tear-strip handle to facilitate unpacking. The metal liners shall be packed in wooden boxes, so as to insure acceptance by common or other carriers, for safe transportation, at the correct rate, to the point of delivery.

b. Reels or coils. -- Unless otherwise specified, the reels or coils of fuse shall be delivered in waterproofed containers so constructed as to insure acceptance by common or other carrier, for safe transportation, at the correct rate, to the point of delivery.

c. Place a data card 5 by 8 inches, made of commercial manila tag board (approximately 200 pounds ream weight) and properly filled out according to the form below, in each packing box liner and container. Two of these cards for each lot shall accompany each shipping ticket, and six of these cards for each lot shall be furnished the inspector.

AMMUNITION DATA CARD

Kind:

Lot No. ----------
Quantity in lot -------
Spec No. -----Rev. ---

Dwg. No. --------Ref. ------
Manufactured by----------
Contract No. ----------Order No. ----------
Packed ----------in a box (carton)
Remarks:

Certified to by ----------
Inspector

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3. Marking.-- a. Packages shall be labeled showing the number of pieces of fuse, grade, limits of time of burning, and lot number.

b. Reels and coils shall be labeled showing the grade, limits of time of burning and lot number.

c. Shipping boxes and containers shall be marked in accordance with Specification No. 100-2, and in addition shall be stamped with the lot number.

VIII NOTES.

1. The fuse specified herein is for use in the manufacture of Livens Projector Fuzes, Hand and Rifle Grenade Fuzes.

2. The use of this specification, whenever applicable, is mandatory on all procuring agencies of the Army.

NOTICE:-- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility or any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any patented invention that may be in any way related thereto.

NOTE.-- Copies of this specification may be obtained upon application to the Office, Chief of Ordnance, Washington, D. C.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

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